

THE INTERN DIARIES

Intern Report

A/B Test Exploratory Analysis

WEEK 3 | DAILY JOURNAL ENTRY



LEARNING BY DOING, ONE DAY AT A TIME

A/B Testing Dataset (TESTING PURPOSE)

1. What is A/B testing?

Answer: A/B testing compares two versions of something using data.

For example: **Control** -> existing version

Variant -> new version

Users are assigned to one of the two groups, and we measure an outcome such as:

Converted = Yes / No

2. What is a Control group?

Answer: The **control** is the baseline against which the new variant is compared.

For example: **Control** -> current website

Variant -> redesigned website

Without a control group, you don't have a proper baseline for determining whether the change itself is associated with difference in conversion.

3. What is the difference between Statistical significance vs practical significance?

Answer: **Statistical significance** asks whether the observed difference provides sufficient evidence against the null hypothesis under the chosen statistical model. **Practical significance** asks whether the size of that difference is meaningful for the business. A result can be statistically significant but too small to justify a costly change, while a potentially valuable difference may fail to reach statistical significance when the sample size is limited.

4. Why use a control group?

Answer: A control group provides a baseline against which the treatment can be compared. Without a control group, it becomes difficult to determine whether the observed conversion level is associated with the new treatment or simply reflects the normal performance of the system. In a properly randomized experiment, comparing the treatment with control helps isolate the effect of the tested change, subject to the experiment's assumptions and design.

5. What does a p-value tell you?

Answer: It quantifies how incompatible the observed result would be with the null hypothesis, under the statistical model and assumptions used. It does not tell me the probability that the null hypothesis is true, nor does it tell me whether the effect is practically important.

6. Why is sample size important?

Sample size affects the precision of estimates and the amount of uncertainty around them. A larger sample can make the conversion rate estimates more precise, while a small sample can produce wide confidence intervals and less decisive evidence.